MIL-G-80162A
8 July 1974
SUPERSEDING
MIL-G-80162
23 September 1968

MILITARY SPECIFICATION

# GRINDING MACHINES, THREAD, EXTERNAL, AUTOMATIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

* 1.1 Scope. This specification covers electric motor driven, floor mounted, horizontal spindle, power operated, automatic thread grinding machines with equipment and accessories as described herein.

* 1.2 Classification. The grinding machines covered by this specification shall be of the following sizes. The size furnished shall be as specified (see 6.2.1):

| Size | Thread Diameter (Inches) | Distance Between Headstock and Tailstock Centers (Inches) |
|---|---|---|
| 5/8 - 6 | 5/8 | 6 |
| 620 | 6 | 20 |
| 636 | 6 | 36 |
| 1245 | 12 | 45 |
| 1248 | 12 | 48 |
| 1284 | 12 | 84 |
| 12144 | 12 | 144 |

** When another size is required, the machine size and its corresponding characteristics shall be as specified under Table I.

## 2. APPLICABLE DOCUMENTS

2.1 Military publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 3415

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

| | |
|---|---|
| MIL-W-17929 | Wheels, Abrasive, Precision |
| MIL-M-18058 | Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of |

STANDARDS

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Marking of U.S. Military Property |
| MIL-STD-461 | Electromagnetic Interference Characteristics Requirements for Equipment |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) OF 1970

| | |
|---|---|
| Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and amendments | Occupational Safety and Health Administration Standards |

NATIONAL BUREAU OF STANDARDS HANDBOOK H28

Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Pringing Office, Washington, DC 20402.)

AMERICAN NATIONAL STANDARDS

| | |
|---|---|
| ANS B5.1 | T-Slots - Their Bolts, Nuts, Tongues and Cutters |
| ANS B5.9 | Spindle Noses |
| ANS B5.10 | Machine Tapers, Self-Holding and Steep Taper Series |
| ANS B5.17 | Markings for Grinding Wheels |
| ANS B7.1 | American Standard Safety Code for the Use, Care, and Protection of Abrasive Wheels |

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American National Standards Institute, Dept. 969, 1430 Broadway, New York, NY 10018.)

JIC ELECTRICAL STANDARDS FOR GENERAL PURPOSE MACHINE TOOLS

* (Application for copies should be addressed to National Machine Tool Builders' Association, 7901 West Park Drive, McLean, VA 22101.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION 360

Manual for Machine Tool Gearing

(Application for copies should be addressed to the American Gear Manufacturers Association, 1330 Massachusetts Ave., NW, Washington, DC 20005.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

## 3. REQUIREMENTS

* 3.1 First Article approval. If First Article approval is required (see 6.2.1 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract.

* 3.2 Design. The machine shall be new and one of the manufacturer's current models. Each machine shall be capable of precision single rib grinding of external right and left hand threads to tolerances specified herein by manual or automatic infeed of the grinding wheel into a traversing and rotating workpiece. The machine capability range in threads per inch shall be manufacturer's standard and shall conform to Table I. When two way grinding is incorporated in the machine design, means to eliminate backlash shall be provided. Unless otherwise specified (see 6.2.1), machines shall be automatic. Each machine shall have provisions for taking multiple or successive grinding passes with a predetermined amount of automatic infeed at the preset time, depending on whether grinding is one-way or two-way. Wheel truing shall be initiated automatically either one or multiple times per grinding cycle, as determined by the machine set-up. The amount of truing feed shall be automatic with compensation for the amount trued off the grinding wheel. Each machine shall complete the grinding of a particular part as determined by the set-up to grind the desired number of passes, true the wheel, compensate, resume the desired



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

number of additional passes and automatically grind to predetermined pitch diameter, without operator's attention. Design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repairs.

3.2.1 Controls. All operating controls shall be grouped in a location convenient to the operator except as otherwise specified herein.

* 3.2.2 Safety and health requirements. Covers or other safety devices shall be provided for all parts of the machine presenting safety hazards which can be guarded without substantial interference to operation of the machine. All guards shall be removable to provide easy access to guarded parts. Except as otherwise specified by the procuring activity, the machine shall comply with the standards promulgated under OSHA applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified (see 6.2.1 and 6.5).

* 3.2.3 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each recirculating system shall include a filter(s) which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

* 3.2.4 Interchangeability. To provide for replacement of worn parts, all parts bearing the same part number shall be functionally interchangeable and shall be dimensionally identical within the manufacturing tolerance limits in use by their manufacturer.

3.3 Construction. The machine shall be so constructed that when installed and connected to power, it will be ready for operation. The machine shall be constructed of parts which are without defect and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing or soldering. Welding, brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently deformed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings and welded parts shall be cleaned and free from sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous material. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's standard commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

3.3.7 Dials. All feed screw dials shall be graduated in the inch system. Graduations to indicate stock removal or wheel movement shall be in increments of not more than 0.001 inch. Dial diameters shall be such that the graduations may be easily read. Dials shall be permanently and legibly engraved, stamped or etched, and shall have a non-glare finish.

3.3.8 Plates. All words on speed and feed indicating plates and on instruction plates shall be in the English language, engraved, etched, embossed or stamped in bold-face characters on a contrasting background.

* 3.3.9 Gears. All gears used in the machine and its components shall be machined in the inch system and shall conform to the applicable requirements of AGMA 360.

3.4 Components. Each machine shall consist primarily of a bed, workslide, headstock, tailstock, grinding wheelhead assembly, wheel truing and dressing device, driving mechanisms and controls.

3.4.1 Bed. The bed shall be ribbed and braced to withstand forces imposed when the machine is utilized to its maximum capacities. Longitudinal guideways shall be provided to accept a traversing workslide. Cross-axis guideways to accept a sliding grinding wheelhead assembly shall be provided. The coolant reservoir, and the hydraulic reservoir when a hydraulic system is utilized, may be housed in the base or provided as separate external units. Means shall be provided to indicate hydraulic fluid level. When applicable, an access door(s) or removable plate(s) shall be provided for servicing components housed within the base.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

3.4.2 Headstock. The headstock shall house the spindle and related components and be constructed to accept, or work in conjunction with, the driving system gear train and workslide to provide the machine capabilities required in 3.2. The headstock may be an integral part of the workslide, or arranged to be longitudinally adjustable along the platen with means for clamping after adjustment. The headstock spindle drive assembly shall be designed to minimize the generation and transmission of vibration. Lubrication for the spindle bearings shall be provided. The headstock spindle drive assembly shall be constructed or adapted to receive a work center, work driving plate or other workholding means, and a driving mechanism. Unless otherwise specified (see 6.2.1), spindle noses and their center mounting surfaces shall be constructed to the machine manufacturer's standards.

* 3.4.3 Tailstock. The tailstock shall house the ram and be longitudinally adjustable along the workslide. Means for positive clamping after adjustment shall be provided. A handwheel, lever or other means shall be provided for adjusting ram movement. The tailstock ram internal taper shall conform to ANS B5.10 and, unless otherwise specified (see 6.2.1), shall be manufacturer's standard. All moving parts shall be shielded against abrasive grit and coolant. The tailstock shall have an adjustable device so that work with a limited amount of taper may be ground, and to facilitate the setting for parallelism.

* 3.4.4 Workslide. The workslide shall be designed to support the headstock, tailstock, and workpiece assemblies and shall be arranged to traverse longitudinally on guideways. Workslide traverse shall be by means of a power operated leadscrew. Adjustable trip dogs or other means shall be provided for limiting traverse. Provisions for longitudinal adjustment of either the headstock or tailstock, or both, when applicable, shall be incorporated. When T-slots are provided, they shall conform to ANS B5.1.

3.4.5 Grinding wheelhead assembly. The grinding wheelhead assembly shall include a wheelhead housing, spindle and its related components, grinding wheel, grinding wheel guards, and when applicable, the truing and dressing device. The spindle drive shall provide operating speeds in accordance with the manufacturer's standard and safe operating speeds of wheels recommended, but not less than 6500 SFPM on minimum safely usable wheel diameter. When crush truing, the assembly shall also provide a low operating speed. The grinding wheel spindle shall be mounted in bearings arranged to minimize axial and radial play. The spindle shall be constructed to accept a driving mechanism at one end and a grinding wheel(s) at the other end. Continuous lubrication to the spindle shall be provided. The assembly shall be mounted on the guideways provided by the bed and shall traverse cross axially to the bed. The assembly shall have both automatic and manual movement for infeed of the grinding wheel. Means shall be provided



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

for engaging the truing and dressing device with the grinding wheel. Means shall be provided for automatic positive size control of the workpiece. The assembly shall be capable of swiveling to the desired helix angle, with means for locking. Graduations shall be provided to indicate swivel setting of the wheelhead. Automatic infeed when furnished shall have means for controlling infeed increments at alternate reversals during one-way grinding and at each reversal during two-way grinding, and for limiting the total infeed during the overall grinding cycle. A handwheel or other manually operated device shall be furnished for feeding the grinding wheel into the work. The hand feed shall have a graduated indicator, in accordance with 3.3.7. Unless otherwise specified (see 6.2.1), the grinding wheelhead shall have power rapid positioning.

3.4.5.1 Automatic cycle. Unless otherwise specified (see 6.2.1), an automatic cycle shall be provided. After set-up for automatic cycle, the operator shall only be required to load the workpiece, start the cycle, unload the workpiece and, when applicable reset the wheelhead for the start of the next cycle. Provisions shall include, but not be limited to, the following functions:

a. One-way grind

(1) Wheel advances at rapid traverse to pre-set depth for first pass, headstock spindle rotation and workslide forward traverse starts, coolant flow starts.

(2) At end of forward grinding pass, wheel withdraws and workslide returns at rapid traverse.

(3) Automatic infeed for next grinding pass, workslide forward traverse starts.

(4) Steps 2 and 3 are repeated until correct size is obtained.

(5) Coolant flow stops, machine comes to automatic stop.

b. Two-way grind

(1) Wheel is positioned to pre-set depth for first pass, headstock spindle rotation and workslide forward traverse starts, coolant flow starts.

(2) At end of forward grinding pass, wheel infeeds for return grinding pass, work rotation and workslide traverse reverse.

(3) At end of return pass wheel infeeds for next forward pass.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

(4) Steps 2 and 3 are repeated until correct size is obtained.

(5) Coolant flow stops, machine comes to automatic stop.

3.4.6 Automatic wheel truing device. Unless otherwise specified, an automatic wheel truing device shall be provided. The device shall be power operated and capable of forming a predetermined straight line thread form into a grinding wheel and maintaining that form, by dressing at intervals. The truing device type shall be as specified by the procuring activity (see 6.2.1).

* 3.4.7 Thread matching device. Unless otherwise specified (see 6.2.1), except on the size 5/8-6 machine, a thread matching device shall be provided to facilitate finish grinding of prethreaded work.

* 3.4.8 Taper grinding. When specified (see 6.2.1), except on a size 5/8-6 machine, means shall be provided for grinding tapered threads. Taper characteristics shall be as specified.

* 3.4.9 Internal grinding unit. Unless otherwise specified (see 6.2.1), except on sizes 5/8-6 and 620 machines, each machine shall be equipped with a power driven internal thread grinding unit. The wheel spindle assembly shall be dynamically balanced and shall rotate in preloaded ball or roller bearings. The spindle shall be a sealed unit. The internal grinding unit shall be designed to use replacement quills.

3.4.10 Driving mechanism. Unless otherwise specified (see 6.2.1), the driving mechanism for the headstock shall be by electric motor or variable speed hydraulic motor and the grinding wheelhead assembly shall be driven by electric motor. When applicable, means shall be provided for adjusting belt tension unless belt tension is maintained automatically.

3.4.11 Grinding wheels. Grinding wheels shall conform to MIL-W-17929. The wheels shall be marked in accordance with ANS B5.17 and shall meet the safety requirements designated in ANS B7.1.

* 3.4.12 Electrical equipment. Unless otherwise specified (see 6.2.1), the machine electrical system shall conform to JIC Electrical Standards for General Purpose Machine Tools, shall accept power from 230/460 volt, 3 phase, 60 hertz systems, and shall be initially wired for operation on 230 volts. An identified terminal shall be provided for grounding the machine when installed. The terminal shall be suitable for connection of the size grounding conductor specified in Table 15-1 of JIC Standards.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

3.4.12.1 Motors. Motors shall be single or multiple speed AC, or variable speed DC, having ball or roller bearings and rated for continuous or intermittent duty. Intermittent duty motors shall be rated not less than 60 minute duty. Each motor used in the machine system shall be suitable for its application, and unless otherwise specified (see 6.2.1), shall meet the requirements of a dripproof enclosure.

3.4.13 Coolant system. A coolant system shall be furnished. It shall include a sump or reservoir, a power driven pump, and piping. The coolant reservoir shall be of sufficient capacity to permit full flow cooling. The system shall include a baffle(s) or strainer which is easily removable for cleaning. The reservoir shall be equipped with means for emptying and cleaning. A coolant delivery nozzle and valve shall be provided to permit the operator to direct and control the amount of coolant flow.

3.4.14 Hydraulic system. When a hydraulic system is provided, it shall be complete including all pumps, valves, piping, cylinders, and pressure controls. Overload protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid to the hydraulic system. The hydraulic reservoir shall have a sight gage to indicate the fluid level.

3.5 Machine characteristics. The machine shall be in accordance with applicable requirements of Table I. Option shall be furnished only when specified.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

TABLE I - SIZES AND CAPACITIES

| CHARACTERISTIC | ** REQUIREMENTS | | | | | | |
|---|---|---|---|---|---|---|---|
| | SIZE 5/8-6 | SIZE 620 | SIZE 636 | SIZE 1245 | SIZE 1248 | SIZE 1284 | SIZE 12144 |
| Capacity | | | | | | | |
| Thread, diameter, inches maximum | 5/8 | 6 | 6-1/2 | 12 | 12 | 12 | 12 |
| Swing, diameter, inches | 10 | 6 | 7-1/2 | 21 | 19 | 19 | 19 |
| Length (between centers), inches | 6 | 20 | 36 | 45 | 48 | 84 | 144 |
| Threaded workpiece length, inches | 3 | 8 | 12 | 18 | 36 | 72 | 120 |
| **Threads per inch obtainable, standard | 10 to 80 | 4 to 80 | 2 to 48 | 2 to 128 | 1 to 128 | 1 to 128 | 1 to 64 |
| Diameter workpiece, insertable through headstock spindle, inches | 17/32 | 0.750 | 2-5/8 | 2-5/8 | 3 | 3 | 6-1/2 |
| Grinding Wheel | | | | | | | |
| Outside diameter (new standard), inches | 18 | 18 | 20 | 20 | 18 | 18 | 18 |
| Outside diameter (minimum safely usable), inches not more than | 14-1/2 | 14 | 16 | 16 | 13 | 13 | 13 |
| Hole diameter, inches | 10 | 9 | 10 | 10 | 9 | 9 | 9 |
| Width range, inches (standard wheel) | 3/8 only | 3/8 to 2 | 3/8 to 2 | 3/8 to 2 | 3/8 to 2 | 3/8 to 2 | 3/8 to 2 |
| Speeds, number | infinite | 10 | infinite | infinite | 10 | 10 | 10 |
| Speed range, RPM | 1340 to 3260 | 1485 to 2640 | 965 to 2350 | 965 to 2350 | 1485 to 2640 | 1485 to 2640 | 1485 to 2640 |
| Grinding Wheelhead | | | | | | | |
| Cross axial adjustment, inches | 4 | 7 | 5 | 8 | 9-1/4 | 9-1/4 | 9-1/4 |
| Rapid traverse movement, inches | 3/8 | 1 | 5/16 | 5/16 | 15/16 | 15/16 | 15/16 |
| Rapid traverse, IPM | 50 | 60 | 5.3 | 5.3 | 60 | 60 | 60 |
| Wheel feed, on work diameter (per pass, inches) | 0.001 to 0.050 | 0.001 to 0.030 | 0.001 to 0.030 | 0.001 to 0.030 | 0.001 to 0.030 | 0.001 to 0.030 | 0.001 to 0.030 |
| Automatic wheelfeed, on work diameter (total) inches | 0.060 | 0.170 | 0.330 | 0.330 | 0.170 | 0.170 | 0.170 |
| Dresser feed, on wheel diameter, range inches | 0.0004 to 0.0006 | 0.0004 to 0.010 | 0.0004 to 0.004 | 0.0004 to 0.004 | 0.0004 to 0.010 | 0.0004 to 0.010 | 0.0004 to 0.010 |
| Helix setting, degrees, left hand | 10 | 25 | 30 | 30 | 45 | 45 | 45 |
| " " right hand | 10 | 25 | 25 | 30 | 45 | 45 | 45 |
| Headstock | | | | | | | |
| Spindle speeds, number | 24 | infinite | 36 | 36 | infinite | infinite | infinite |
| Spindle speed range RPM | 30 to 712 | 4 to 120 | 0.6 to 150 | 0.36 to 94 | 1/2 to 160 | 1/2 to 160 | 1/2 to 160 |
| Optional RPM | - | - | - | - | - | - | - |
| Motors, Horsepower | | | | | | | |
| Wheelhead | 5 | 3 | 7-1/2 | 7-1/2 | 3 | 3 | 3 |
| Headstock | 3/4 | 1 | 1-1/2 | 1-1/2 | 2 | 2 | 2 |
| Coolant | 1/2 | 3/4 | 3/4 | 3/4 | 3/4 | 3/4 | 1-1/2 |
| Hydraulic | none | 1 | none | 3 | 3 | 3 | 3 |

** Unless otherwise specified (see 6.2.1).

Requirements are not less than those shown, except as otherwise indicated above. Where a range is shown, the requirement is from the lower figure or less to the larger figure or more.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

3.6 Performance. Each machine shall be capable of grinding workpieces in accordance with the requirements herein, and specifically in accordance with the test requirements designated in 4.5.2.

3.7 Alignment tolerances. Each machine shall meet the alignment tolerances of 4.5.3.

* 3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of wrenches normally furnished by the manufacturer.

b. A total of three (3) grinding wheels, type, dimensional characteristics and composition to be as specified (see 6.2.1), complete with wheel mount and balancing weights, as required.

c. One wheel mount balancing arbor.

d. One extra set headstock and wheelhead driving belts, if applicable.

e. Except on sizes 5/8-6, and 620, one slotted face plate.

f. Except on size 5/8-6, one independent 4-jaw chuck, size to be compatible with machine.

g. One set of workdriving dogs, hole size as specified (see 6.2.1).

h. One work area enclosure, and vapor exhaust system.

i. Two work centers.

j. One wheel sleeve, and one wheel sleeve puller, if required.

3.9 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2.1).

* 3.10 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461.

3.11 Fungus control. When required (see 6.2.1), fungus proofing shall be as specified.

3.12 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a trans-



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

parent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application
Service interval
Type of lubricant
Viscosity
Federal or military specification number (if available)

3.13 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate identifying the machine in accordance with MIL-STD-130 shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each such caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Size_____
Thread diameter_____inches
Center to center distance between headstock and tailstock_____ inches
Power input (volts, total amps, phases, cycles)
Contract Number or Order Number
Federal Stock Number or Plant Equipment Code
Date of manufacture
U S

3.14 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.15 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First Article inspection. When First Article approval is required under 3.1, First Article inspection shall be performed. Unless otherwise specified (see 6.2.1), First Article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure to pass the examination or any test shall be cause for disapproval of First Article.

4.3 Quality conformance inspection. Each item shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5 through 4.5.3 and the inspection in 4.6. Failure of the item to pass any examination or test shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined for compliance with the requirements in 3.2 through 3.5 and 3.8 through 3.15.

4.5 Tests.

4.5.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms and other accessories shall be verified during the trial period.

4.5.2 Performance tests. Each machine shall thread grind a workpiece to the maximum thread length for the size machine furnished. The thread angle shall be within $\pm$ 5 minutes on the half angle, and lead within $\pm$ .0002" per inch with accumulative error not to exceed 0.001" in 12". Each automatic machine shall automatically grind threads to a pitch diameter tolerance of $\pm$ 0.0005". The workpiece size, materials and methods used to achieve the above designated requirements shall be at the supplier's option.

4.5.3 Accuracy tests. Each machine and its components shall be tested for conformance with the accuracies specified in Table II. The instruments and methods used in performing the tests shall be at the option of the supplier provided they accurately determine the tolerances of the machine.

4.5.4 Electromagnetic interference control. Unless otherwise specified (see 6.2.1), equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.10.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

TABLE II - ACCURACY TOLERANCES

| TEST TO BE APPLIED | PERMISSIBLE ERROR |
|---|---|
| **Bed and Workslide:** | |
| Bed straight or level in longitudinal direction | 0 to 0.0005" per 12" |
| Bed flat or level in transverse direction no twist permitted | 0 to 0.0005" per 12" |
| Workslide top surfaces flat | 0 to 0.0005" per 12" |
| **Headstock and Spindle:** | |
| Axis of headstock parallel with workslide in vertical plane | 0 to 0.0012" per 12" |
| Axis of headstock parallel with workslide in horizontal plane | 0 to 0.0012" per 12" |
| Center point for true running | 0.0002" TIR |
| Taper of work spindle runs true: | |
| (1) Nearest spindle nose | 0.0002" TIR |
| (2) At a distance of 12 inches | 0.001" TIR |
| **Tailstock:** | |
| Sleeve parallel with workslide in vertical plane | 0 to 0.0012" per 12" |
| Sleeve parallel with workslide in horizontal plane | 0 to 0.0012" per 12" |
| Taper in sleeve parallel with workslide in vertical plane | 0 to 0.0015" per 12" |
| Taper in sleeve parallel with workslide in horizontal plane | 0 to 0.0015" per 12" |
| Axis of centers (mandrel between centers) parallel with workslide in vertical plane | 0 to 0.002" per 12" |
| **Grinding Wheel Spindle:** | |
| Taper of spindle running true | 0.0002" TIR |
| Spindle for axial slip measured at 2 points, displaced in 180° | 0 to 0.001" |
| Axis of wheel spindle parallel with workslide in vertical plane (rising toward free end of spindle) | 0 to 0.0012" per 12" |
| Axis of wheel spindle parallel with workslide in horizontal plane (free end of spindle inclined toward table) | 0 to 0.0012" per 12" |
| Axis of wheel spindle and headstock spindle at same height with respect to workslide | 0 to 0.010" |
| Rise and fall of wheel spindle in its infeed movement | 0 to 0.0015" in total length of infeed motion |
| Infeed motion of wheel slide square with bed ways | 0 to 0.0005" in total length of infeed motion |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

4.6 Inspection of preparation for delivery. Preservation-packaging, packing and marking shall be inspected to determine compliance with the requirements of Section 5.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-M-18058, at Level A or C as specified (see 6.2.1).

5.2 Packing. Packing shall be in accordance with MIL-M-18058, at Level A, B, or C as specified (see 6.2.1).

5.3 Marking. Marking shall comply with MIL-STD-129. Additional marking shall be as specified (see 6.2.1).

## 6. NOTES

6.1 Intended use. The grinding machines covered by this specification are intended for use in shops where a fast, efficient and economical way to traverse grind precision threads and worms is required.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number and date of this specification.

b. Size required (see 1.2).

c. First Article approval, if required (see 3.1).

d. Specify if machine is not to be automatic (see 3.2).

e. Specify exceptions and additional safety and health requirements (see 3.2.2 and 6.5).

f. Painting, if different (see 3.3.5).

g. Headstock spindle characteristics, if different (see 3.4.2).

h. Tailstock ram characteristics, if different (see 3.4.3).

i. Power rapid positioning if not required (see 3.4.5).



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

j. Automatic cycle if not required (see 3.4.5.1).

k. Specify type of automatic truing device required (see 3.4.6).

l. Thread matching device, if not required (see 3.4.7).

m. Specify if taper thread grinding is required, and taper characteristics required (see 3.4.8).

n. Internal grinding unit if not required (see 3.4.9).

o. Driving mechanisms, if different (see 3.4.10).

p. Electrical equipment, if different, and voltage for initial wiring, if different (see 3.4.12).

q. Type of enclosure for motors, if different (see 3.4.12.1).

r. Machine characterisitcs, if different, threads per inch, if different, and option, if required (see Table I).

s. Standard equipment, if different (see 3.8).

t. Grinding wheels: type, dimensions, and composition required (see 3.8(b)).

u. Specify hole size for workdriving dogs (see 3.8(g)).

v. Optional equipment, as required (see 3.9).

w. Electromagnetic interference control, if required (see 3.10).

x. Type fungus proofing, if required (see 3.11).

y. Lubrication chart, if different (see 3.12).

z. Nameplate, if different (see 3.13).

aa. First Article inspection, if different (see 4.2).

bb. Quality conformance inspection, if different (see 4.3).

cc. Electromagnetic interference control test requirements, if different (see 4.5.4 and 6.4).



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

dd. Required level of preservation-packaging (see 5.1).

ee. Required level of packing (see 5.2).

ff. Additional marking, if required (see 5.3).

6.2.2 Contract data requirements. Required technical data, such as operator's manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

6.3 First Article. When First Article approval is required under 3.1, the solicitation should adhere to the applicable portions of Section 1, Part 19 of the Armed Services Procurement Regulation.

6.4 Electromagnetic testing. When equipment requires electromagnetic interference control testing, such testing should be specified to be in accordance with MIL-STD-461.

6.5 Safety and health determinations. In order that equipment integrated into the user's operational environment will comply with the standards promulgated under the Occupational Safety and Health Act of 1970, limitation and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc. as applicable, specific requirements concerning guarding the point(s) of operation, and other safety and health requirements should be specified.

6.6 Size identification. Machines covered by this specification are sized by the maximum thread diameter and the center distance between headstock and tailstock.

* 6.7 Size changes. In this revision, Size 5/8-4 has been increased to 5/8-6, although there is no change in the machine involved. Sizes 324 and 410 have been deleted, since they are no longer manufactured.

6.8 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

Custodians:

Army - WC
Navy - SH
Air Force - 84

Review Activities:

Army - SM
Navy - SH
DSA - GS

User Activities:

Army - WC
Navy - AS

Preparing Activity:

DSA-IP

Project Number:

3415-0100



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD 304

HEADQUARTERS
DEFENSE INDUSTRIAL PLANT EQUIPMENT CENTER
MEMPHIS, TENNESSEE ~~38102~~ 38114

DEFENSE SUPPLY AGENCY
OFFICIAL BUSINESS

Commander
Defense Industrial Plant Equipment Center
ATTN: DIPEC-TP
Memphis, Tennessee ~~38102~~ 38114

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.

# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:42Z
Check the source to verify that this is the current version before use.